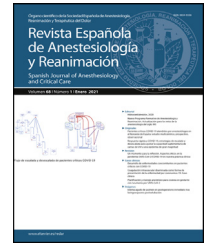




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REVIEW

Adding intrathecal midazolam to local anesthetics improves sensory and motor block and reduces pain score without increasing side effects in lower limb surgeries: A meta-analysis and systematic review



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KEYWORDS

Intrathecal midazolam;
Pain score;
Lower limb surgery;
Opioids consumption;
Sensory block;
Motor block

Abstract This meta-analysis was done to investigate the role of intrathecal midazolam in lower limb surgeries regarding prolongation of spinal block, postoperative pain control and associated side effects. The included studies reported onset and duration of sensory and motor block, time to first request analgesia, 24 h opioid consumption, postoperative pain control, and associated side effects following use of intrathecal midazolam for lower limb surgeries. This review was performed following the PRISMA guidelines and using the online databases, Medline, Science Direct, Google scholar and Cochrane library. We registered this review with the PROSPERO database (ID-CRD42022346361) in August 2022. A total of 10 randomised controlled trials were included in this meta-analysis.

Our results showed patients receiving 1 mg intrathecal midazolam showed significantly faster onset of sensory block [$P = .001$ (CI: $-0.98, -0.31$)]. Duration of sensory and motor block were also significantly prolonged in intrathecal midazolam group [$P < .00001$ (CI: $18.08, 39.12$), $P = .002$ (CI: $0.45, 2$)]. Intrathecal midazolam also increased the time to first request analgesia [$P = .0003$, (CI: $1.22, 4.14$)]. Pain scores at 4 and 12 h postoperatively were significantly lower in patients receiving intrathecal midazolam [$P = .00001$ (CI: $-1.20, -0.47$) and $P = .05$ (CI: $-0.52, -0.01$) respectively]. In conclusion, the addition of intrathecal midazolam to local anesthetics in lower limb surgeries results in early onset of sensory and motor block. It also increases the duration of sensory and motor block. The time to first request analgesia is increased. VAS pain scores at 4 and 12 h postoperatively were also lower without any increased side effects.

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PALABRAS CLAVE

Midazolam intratecal;
Puntuación del dolor;
Cirugía de miembros inferiores;
Consumo de opioides;
Bloqueo sensorial;
Bloqueo motor

La adición de midazolam intratecal al anestésico local mejora los bloqueos sensorial y motor y reduce la puntuación del dolor sin incrementar los efectos secundarios en la cirugía de miembros inferiores. Metaanálisis y revisión sistemática

Resumen Este metaanálisis investiga el impacto de midazolam intratecal en la anestesia espinal, el control del dolor postoperatorio y los efectos secundarios relacionados con la anestesia en la cirugía de miembros inferiores. Realizamos una búsqueda en Medline, Science Direct, Google Scholar y Cochrane Library de los estudios que reportaron el inicio y la duración de los bloqueos sensorial y motor, el tiempo transcurrido hasta la primera solicitud de analgesia, el consumo de opioides durante 24 horas, el control del dolor postoperatorio, y los efectos secundarios tras la administración de midazolam intratecal en pacientes sometidos a cirugía de miembros inferiores. Se identificaron diez estudios, que se incluyeron en el metaanálisis. La revisión fue realizada siguiendo las directrices PRISMA, registrándose en la base de datos PROSPERO (ID-CRD42022346361) en agosto de 2022.

Nuestros resultados muestran que los pacientes que reciben 1 mg de midazolam intratecal reflejaron un tiempo de inicio de bloqueo significativamente más alto ($P = ,001$ (IC: $-0,98, -0,31$)), mayor duración de los bloqueos sensorial y motor [$P < ,00001$ (IC: $18,08, 39,12$), $P = ,002$ (IC: $0,45, 2$), y mayor tiempo transcurrido hasta la primera solicitud de analgesia de rescate [$P = ,0003$, (IC: $1,22, 4,14$)]. Las puntuaciones de dolor a las 4 y 12 horas postoperatorias fueron significativamente inferiores en los pacientes que recibieron midazolam intratecal [$P = ,00001$ (IC: $-1,20, -0,47$) y $P = ,05$ (IC: $-0,52, -0,01$) respectivamente]. En conclusión, la adición de midazolam intratecal al anestésico local en la cirugía de miembros inferiores acorta el tiempo de inicio de los bloqueos sensorial y motor, incrementa la duración del bloqueo, y prolonga el tiempo transcurrido hasta la primera solicitud de analgesia. Las puntuaciones del dolor a las 4 y 12 horas postoperatorias fueron menores, no observándose efectos secundarios adicionales. © 2023 Publicado por Elsevier España, S.L.U. en nombre de Sociedad Española de Anestesiología, Reanimación y Terapéutica del Dolor.

Introduction

Lower limb surgeries are one of the most important and widely performed surgeries nowadays. Considering the simplicity and reliability of spinal anesthesia, it has become one of the preferred techniques in these surgeries. Although, early regression of block unexpectedly or prolonged operative time can sometimes result in pain intraoperatively or during early postoperative period.¹ Inadequacy in postoperative pain control has adverse implications on patient's physiological, metabolic and psychological condition.² Increased dose of local anesthesia or use of additives can be helpful to improve duration and quality of spinal anesthesia. However, higher local anesthesia doses can result in significant perioperative hypotension which increases the risk of cardiac and neurological complications especially in vulnerable patients.³

Several intrathecal additives have been used to improve speed of onset, prolong duration of block and better quality of analgesia. The most commonly used additives include opioids but these are associated with side effects including nausea and vomiting, respiratory depression, pruritus and urinary retention.⁴ Other agents have also been used intrathecally which include ketamine, clonidine, neostigmine, vasoconstrictors, sodium bicarbonate and benzodiazepines etc.⁵

In anesthesia practice, benzodiazepines are mainly utilized for anxiolysis, amnesia and sedation. But many studies including randomised controlled trials have been done which have investigated the role of intrathecal benzodiazepine like midazolam in combination with local anesthetics. We could only find two meta-analyses on the role of intrathecal midazolam in scientific literature search. The first one was done in 2008 and included mainly obstetric and general surgery patients.⁶ The second one was done very recently in 2022 and it included only patients undergoing caesarean sections.⁷ The aim of this systematic review and meta-analysis was to investigate the role of intrathecal midazolam in lower limb surgeries regarding prolongation of spinal block, postoperative pain control and associated side effects.

Material and methods

This systematic review and meta-analysis were conducted in July 2022 and search was repeated in August 2022 in order to ensure accuracy. We registered the review with International Prospective Register of Systematic Reviews (PROSPERO) as ID-CRD42022346361 in July 2022. We did the search and review in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.⁸

Search strategy and inclusion criteria

The search strategy was developed for PubMed, Science Direct, Cochrane library and Google Scholar. It comprised randomised controlled trials (RCTs) published in English and related to patients who received intrathecal midazolam for lower limb surgeries. The primary outcomes were onset of sensory and motor block and duration of sensory and motor block. The secondary outcomes were time to first request analgesia, 24 h pain control and occurrence of side effects including hypotension, bradycardia, postoperative nausea and vomiting (PONV) and urinary retention. We considered only primary research for our review while abstracts, comments, review articles, and technique articles were excluded.

Data extraction

We extracted data using a pre-designed proforma regarding populations and outcomes from individual studies. It included information like studies' general details (Journal, year of publication, design, groups, and outcomes), study participants, sample size, intervention (dosages and timing of administration) and outcomes (onset and duration of block, time to first request analgesia, pain control and side effects). Means and standard deviations of continuous data were extracted from the accompanying tables or graphs.

The search was performed by two researchers independently. During the entire process, all discrepancies were settled through discussion between the researchers. Both authors independently appraised the individual studies according to the Consolidated Standards of Reporting Trials (CONSORT) checklist.⁹

Statistical analysis

The quality of included studies was assessed using risk of bias 2 tool (RoB2).²⁰ For meta-analysis of the included studies, Review Manager (RevMan for Mac, version 5.4; Cochrane Collaboration, Oxford, UK) was used. The heterogeneity of data was assessed by measuring I^2 . Data on onset and duration of sensory and motor block were pooled. Similarly, data on time to first request analgesia and pain control at different time points were pooled. Side-effects were also pooled with no time points. For continuous data, the mean difference or standardized mean difference were used to report the treatment effect, while odds ratios were used for dichotomous data. A random-effect model was used for the meta-analysis due to expected heterogeneity in the studies. We expected higher heterogeneity because of different timings and dosages of intravenous ketamine used and also difference in primary outcomes in the included studies. $P \leq .05$ was set as the level of statistical significance.

Results

We found 664 studies using our search strategy including 'intrathecal midazolam' OR 'spinal midazolam' AND 'lower limb surgeries' in PubMed, Science Direct, Google scholar and Cochrane library. After thoroughly assessing as demon-

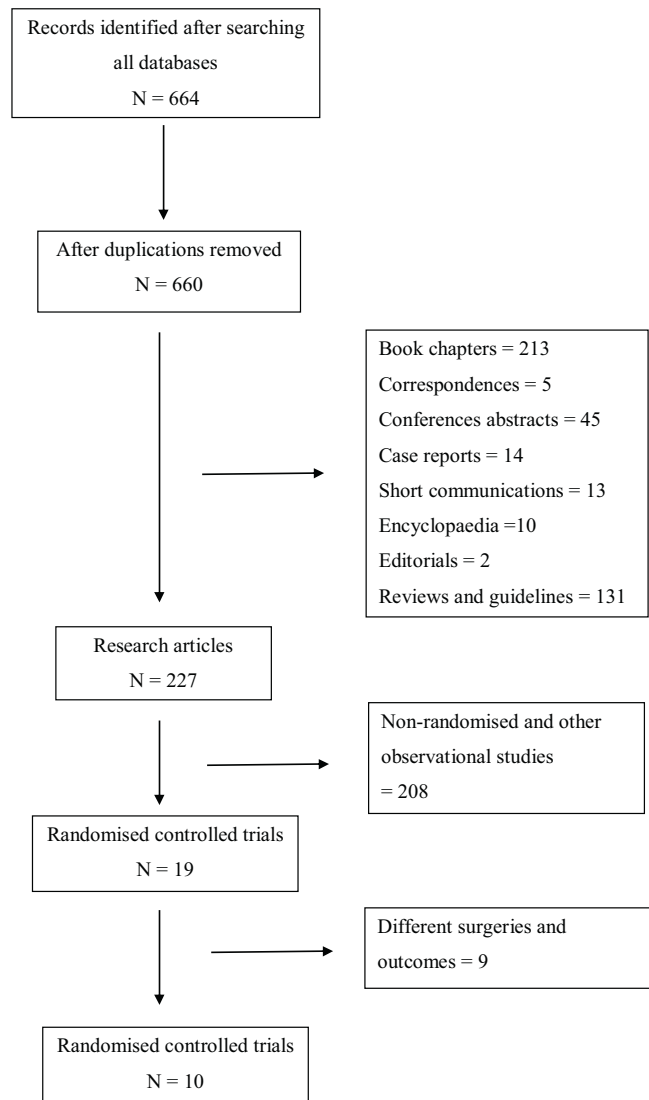


Figure 1 Flow diagram of search process.

strated in Fig. 1, we identified 10 studies to be included in our review. RoB-2 tool assessed all the studies to have low risk of bias.

Onset of sensory block

A pooled meta-analysis of included studies did not show any statistical difference in time of onset of sensory block between control group vs midazolam group by 17 s (mean difference -0.28) as demonstrated in Fig. 2a ($P = .10$). Sub-group pooled analysis of patients receiving 1 mg intrathecal midazolam showed significantly faster onset of sensory block by 39 s (mean difference -0.65) as demonstrated in Fig. 2b.

Onset of motor block

Pooled meta-analysis of included studies found statistical difference in time of onset of motor block in intrathecal midazolam group when compared to control group by

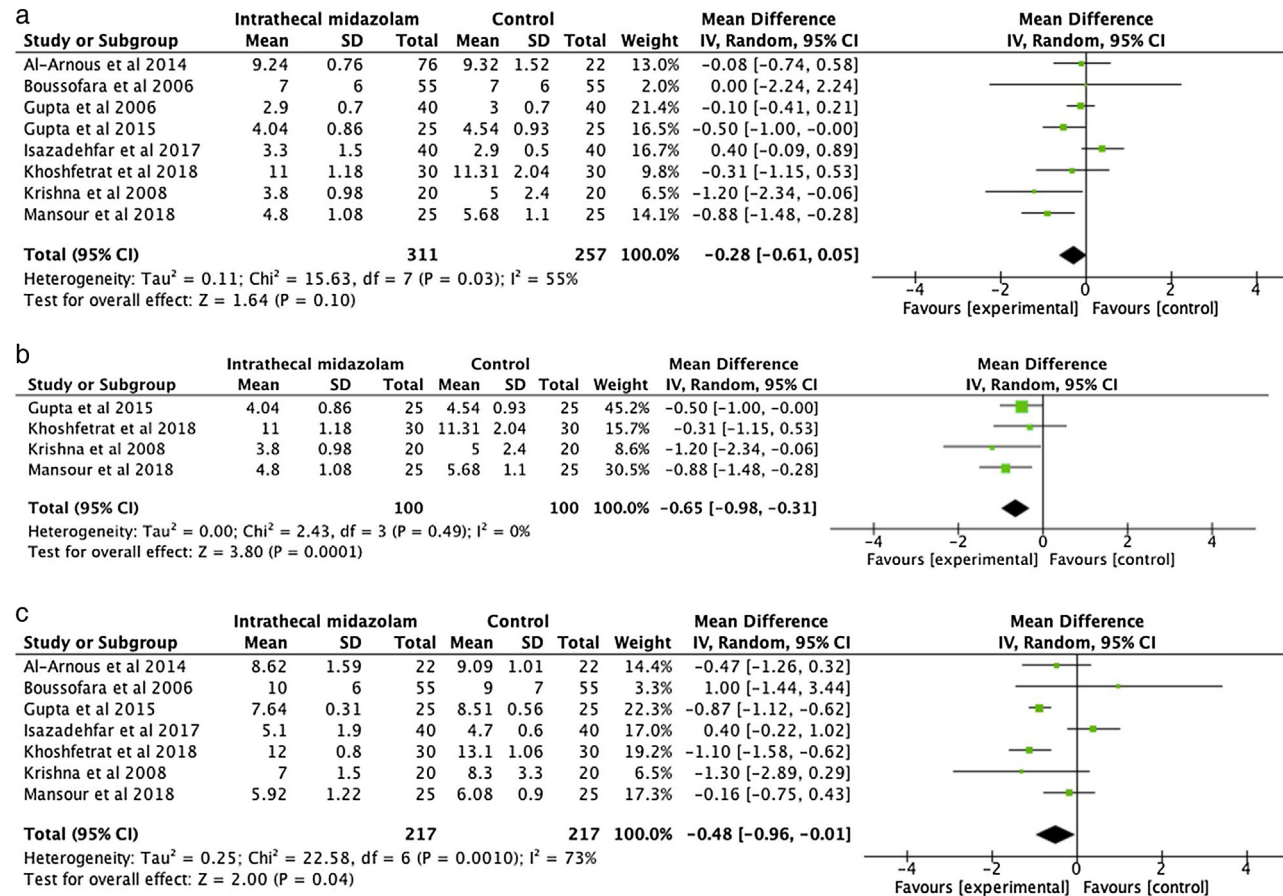


Figure 2 Characteristics of sensory and motor block. **a:** Time of onset (in seconds) of sensory block. **b:** Time of onset (in seconds) of sensory block after 1 mg intrathecal midazolam. **c:** Time of onset (in seconds) of motor block.

29 s (mean difference -0.48) as demonstrated in Fig. 2c ($P = .04$).

Duration of sensory block

Pooled meta-analysis of all included studies found highly significant difference in duration of sensory block between control group and intrathecal midazolam group by mean difference of 28.6 min as demonstrated in Fig. 3a ($P < .00001$).

Duration of motor block

Pooled meta-analysis of included studies showed a significant difference in duration of motor block with intrathecal midazolam in comparison to control group by mean difference of 23.85 min as demonstrated in Fig. 3b ($P = .002$).

Time to first request analgesia

Pooled meta-analysis of the included studies showed a highly significant difference in time to first request analgesia in the group which received intrathecal midazolam compared to control group by 118.25 min as demonstrated in Fig. 4 ($P = .0003$).

Pain scores

Pooled meta-analysis of included studies found significantly different visual analogue scale (VAS) pain scores between control group vs intrathecal midazolam group at 4 and 12 h postoperatively by 0.84 and 0.80 respectively as demonstrated in Figs. 5a and 5 b ($P = .00001$, $P = .05$).

Side effects

Hypotension

Pooled meta-analysis of included studies did not demonstrate any significant difference in the occurrence of hypotension between control vs midazolam group with risk ratio, 0.94 and odds ratio, 0.92 ($P = .79$).

Bradycardia

Pooled meta-analysis of included studies did not demonstrate any significant difference in the occurrence of bradycardia between control vs midazolam group with risk ratio, 1 and odds ratio, 1 ($P = .99$).

Postoperative nausea and vomiting (PONV)

Pooled meta-analysis of the included studies did not show any significant difference in occurrence of PONV by adding intrathecal midazolam in patients undergoing lower limb surgeries with risk ratio, 1.79 and odds ratio, 1.89 ($P = .15$).

Discussion

The current meta-analysis showed that use of intrathecal midazolam as an additive to local anesthesia with or without opioids resulted in prolonged sensory and motor block. Use of 1 mg intrathecal midazolam resulted

in faster onset of sensory block. It also enhanced the time to first request analgesia. Postoperative pain scores were lower in midazolam group at 4 and 12 h postoperatively. There was no difference in the occurrence of associated side effects when midazolam was added intrathecally.

GABA (Gamma aminobutyric acid) receptors are found in spinal cord and play an important role in anti-nociception. Intrathecal midazolam reduces the GABA mediated neurotransmission at the interneuron level resulting in decreased dorsal horn neuron excitability. Additionally, midazolam induces release of endogenous opioids which act on delta receptors in spinal cord. As a result, intrathecal midazolam improves quality of intraoperative anesthesia and improves analgesia postoperatively.^{13,21}

Our meta-analysis showed that intrathecal midazolam did not result in faster onset of sensory block although motor block was found to be more rapid by 29 s. Although subgroup meta-analysis of 1 mg intrathecal midazolam resulted in faster onset of sensory block as well by 39 s. Amin et al.²² found that the addition of intrathecal midazolam resulted in faster onset of both sensory and motor block in patients undergoing caesarean section. In another study, Laskar et al.²³ demonstrated a faster sensory-motor block by addition of midazolam to intrathecal bupivacaine. Agrawal et al. showed a faster onset of only sensory block in midazolam group.²⁴ In a recent meta-analysis on patients undergoing caesarean section,⁷ onset of both sensory and motor block was found to be faster with the addition of intrathecal midazolam.

Hung et al.⁷ showed a longer sensory block by using intrathecal midazolam with mean difference of 39.76 min ($P = .001$). This result is consistent with our meta-analysis which demonstrated a mean difference of 28.6 min in duration of sensory block with intrathecal midazolam. Indrajit et al.²⁵ and Malavika et al.²⁶ also found a relatively longer sensory block with intrathecal midazolam. Yegin et al.²⁷ on the other hand did not find any significant increase in duration of sensory block.

Our meta-analysis demonstrated a prolonged motor block with mean difference of 23.85 min in patients who received intrathecal midazolam. This was in contrast to findings by Hung et al.⁷ which did not show any significant increase in duration of motor block. Another previous meta-analysis by Ho et al.⁶ also did not find any significant difference in duration of motor block between control group and intrathecal midazolam group. Similar findings were also found in the study by Shadangi et al.²⁸ On the other hand, Amin et al.²² and Bharti et al.²⁹ reported a significantly increased duration of motor block in patients who received intrathecal midazolam. Chattopadhyay et al.³⁰ also demonstrated a significantly increased duration of motor block (225 min vs 195 min) in patients receiving intrathecal midazolam. Clinically, this increase in duration of sensory and motor block could be of major clinical significance as this will help in avoiding spinal anesthesia conversion to general anesthesia for surgeries which unexpectedly get prolonged. A recently published study³¹ demonstrated that overall failure rate of spinal block in different surgeries was 16.6% and the most common reason for failure was the inadequate spinal block to initiate or complete the surgery. Considering the effect of intrathecal midazolam in faster onset of sensory block and

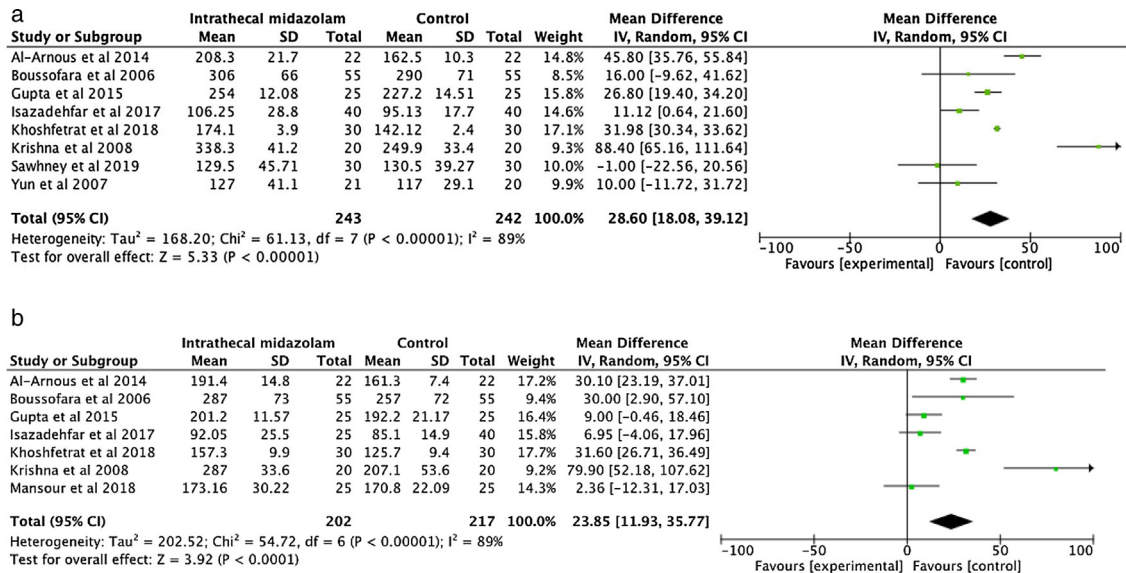


Figure 3 Duration of sensory and motor block. **a:** Duration of sensory block (in minutes). **b:** Duration of motor block (in minutes).

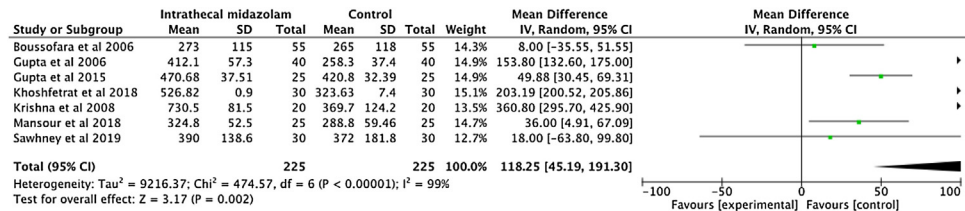


Figure 4 Time (in minutes) to first request analgesia.

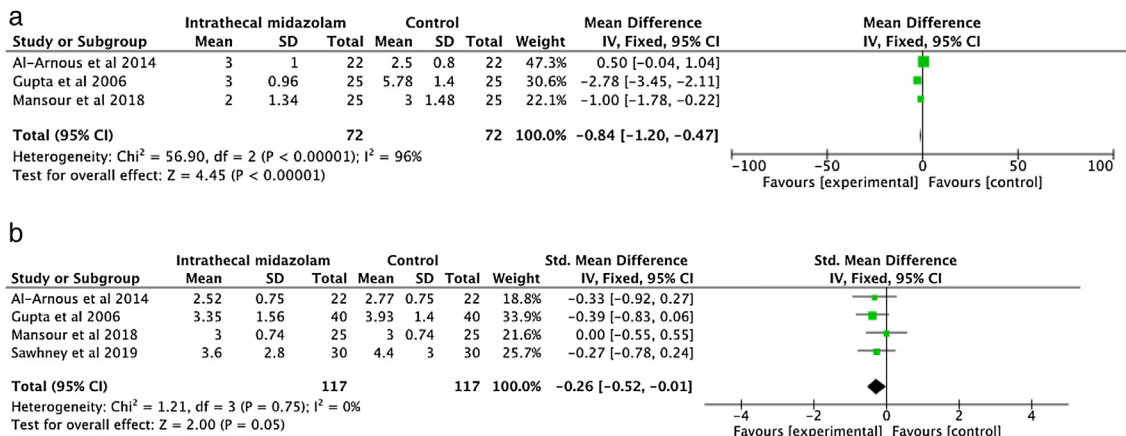


Figure 5 VAS Pain scores postoperatively. **a:** VAS Pain score at 4 h postoperatively. **b:** VAS Pain score at 12 h postoperatively.

increasing duration of sensory and motor blocks by 28.6 and 23.85 min respectively in our study, this seems to be very relevant clinically. We therefore can improve the quality of spinal anesthesia success in our anesthesia practice by using intrathecal midazolam as an additive.

The time to first request analgesia in our meta-analysis was also significantly increased in intrathecal midazolam group with mean difference of 118.25 min. There was no significant difference in time to first request analgesia between control group and intrathecal midazolam group in

the study by Boussofara et al.¹² Gupta et al.¹³ found significant difference in duration of analgesia with addition of intrathecal midazolam. Adding intrathecal midazolam significantly increased the time to first request analgesia in the study by Gupta et al.¹³ Khoshfetrat et al.¹⁵ also found significant difference in duration of analgesia between intrathecal midazolam group compared to another group. Murali et al.¹⁶ demonstrated significantly longer pain free interval in intrathecal midazolam group. In the study by Mansour et al.,¹⁷ a significant difference in time to first

request analgesia was found between two groups. Although no significant difference in duration of analgesia was shown in the study by Sawhney et al.¹⁸ Our results from meta-analysis are consistent with results from meta-analysis by Hung et al.⁷ with relatively lesser increase in duration (MD: 59.96 min). Ho et al.⁶ also demonstrated a longer time to first request analgesia with mean difference of 98.7 min in midazolam group ($P < .00001$). Amin et al.²² in their study also found significantly prolonged effective analgesia time in midazolam group (224.2 ± 18.6 min) compared to control group (185.5 ± 17.45 min).

Our meta-analysis showed that postoperative VAS pain score was significantly lower at 4 and 12 h after surgery in intrathecal midazolam group. Similarly, Ho et al.⁶ demonstrated an improved pain scores at 2 and 4 h with mean difference as 0.98 ($P = .001$). Lesser pain scores in postoperative period were also noted in studies by Alam et al.³² and Codero et al.³³ Previously, different studies have used different values as clinically significant pain reduction. Myles et al.³⁴ demonstrated that a change of 10 in 100 mm VAS signifies as clinically significant improvement or deterioration. Danoff et al.³⁵ showed a pain score change of 18.6 and 22.6 in 100 mm VAS pain scale for total hip arthroplasty and total knee arthroplasty respectively as clinically meaningful pain improvement. Considering these thresholds, our VAS score changes of 0.84 and 0.80 on 10 cm VAS scale are somewhat below the meaningful values in literature.

Scientific literature postulates the role of midazolam in preventing PONV. The possible mechanism is by activation of GABA receptors in chemoreceptor trigger zone.³⁶ Ho et al.⁶ reported that intrathecal midazolam decreased PONV by 50% (OR 0.50, CI 0.27–0.90, $P = .02$). Similarly, Hung et al.⁷ showed that patients in intrathecal midazolam group had significantly lesser PONV events (OR 0.28, CI 0.17–0.45, $P < .00001$). Chattopadhyay et al.³⁰ also demonstrated a lower incidence of PONV in patients receiving intrathecal midazolam. These findings are in contrast to our meta-analysis which did not show any significant difference in PONV with addition of intrathecal midazolam in patients undergoing lower limb surgeries. Amin et al.²² in their study also mentioned similar findings with no significant difference in PONV.

Our meta-analysis did not show any significant difference in occurrence of hemodynamic events like hypotension and bradycardia with use of intrathecal midazolam. The finding is consistent with the results in recent meta-analysis on patients undergoing caesarean section.⁷ There was no difference between two groups regarding other side effects like sedation, pruritus, respiratory depression and urinary retention which is consistent with previous meta-analyses.^{6,7}

There are few limitations in our study. Firstly, although in all included studies, we used the data from two groups; one with intrathecal midazolam and other without intrathecal midazolam. But some studies used other additives as well. In study by Boussofara et al.,¹² one group received intrathecal bupivacaine with clonidine and second group received midazolam along with the other two. Khoshfetrat et al.¹⁵ compared three groups in their study. One group received only intrathecal bupivacaine; second group received bupivacaine and fentanyl while third group received midazolam along with bupivacaine and fentanyl. Mansour et al.¹⁷ compared three groups in their study. First group received

intrathecal bupivacaine, second group received bupivacaine and dexmedetomidine while third group received bupivacaine, dexmedetomidine and midazolam. In the study by Murali et al.,¹⁶ three groups were compared. First group received intrathecal bupivacaine, second group received ketamine additionally, while third group received bupivacaine, ketamine along with midazolam. While in other studies by Al-Arnous et al.,¹⁰ Gupta et al.,¹¹ Gupta et al.,¹³ Isazadehfar et al.,¹⁴ Khoshfetrat et al.,¹⁵ Sawhney et al.¹⁸ and Yun et al.,¹⁹ we used data from two groups without any other additives. Another limitation was that the dosages of intrathecal midazolam differ in different studies. Five out of 10 studies used 1 mg intrathecal midazolam and 4 studies used 2 mg or more intrathecally, while one study used 0.02 mg/kg intrathecal midazolam. The dosage of bupivacaine used for spinal also varied from 7.5 mg upto 17.5 mg. Another limitation is that all the studies did not report any delays in discharge or re-admission due to poor pain control or any other complication. Also, the risks of developing long term complication of intrathecal midazolam could not be assessed with this meta-analysis. In conclusion, the addition of intrathecal midazolam to local anesthetics in spinal anesthesia for lower limb surgeries results in early onset of sensory and motor block. It increases the duration of sensory and motor block. The time to first request analgesia is also increased. Pain score at 4 and 12 h postoperatively was lower without any increased side effects.

Disclosures

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Acknowledgment

No.

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